Commutative Algebra Problem Set #2

Dmytro Zakharov

Problem (Exercise 1)**.** *Consider the following rings:*

$$A_1 = \mathbb{C}[x] \qquad A_2 = \mathbb{C}[x, x^{-1}] \qquad A_3 = \mathbb{C}[x^2, x^3]$$
$$A_4 = \mathbb{C}[x, y] \qquad A_5 = \mathbb{C}[x]/\langle x^6 \rangle \qquad A_6 = \mathbb{C}[x, y]/\langle x^2, y^3 \rangle$$
$$A_7 = \mathbb{C}[x, y]/\langle xy \rangle \qquad A_8 = \mathbb{C}[x, y]/\langle xy^2 \rangle \qquad A_9 = \mathbb{C}[x, y]/\langle xy, y^2 \rangle$$

Prove that the rings A_i and A_j are not isomorphic as $\mathbb{C}$-algebras for $i \neq j$. Can you show they are also not isomorphic as rings?

Solution.

A_2. Now let us look at A_2. It is clearly not isomorphic to the rest due to the nature of invertible elements $A_2^\times$. Indeed, $A_2^\times$ consits of all elements of form cx^n for $c \in \mathbb{C}$ and $n \in \mathbb{Z}$. All other rings have units in $\mathbb{C}^\times$ only.

A_3. This ring consists of all finite sums $\sum_i \sum_j c_{i,j} x^{2i+3j}$ for $i, j \in \mathbb{Z}_{\geq 0}$. Note that every integer $n \neq 1$ can be written as $n = 2i + 3j$ for some $i, j \in \mathbb{Z}_{\geq 0}$[*]. So we conclude that A_3 consists of polynomials $\sum_j a_j x^j$ with $a_1 = 0$. What are some properties of A_3 that are not present, say, in $\mathbb{C}[x]$? First, it is not a UFD: for instance, x^6 has two distinct decompositions: $(x^2)^3$ and $(x^3)^2$. In particular, it is not a PID (since UFD implies PID). Moreover, $A_3 \cong \mathbb{C}[x, y]/\langle y^2 - x^3 \rangle$ (where isomorphism is given by $\varphi : x \mapsto x^2, y \mapsto x^3$)[†], and the latter gives a slighly more interesting intuition of A_3. For instance, the curve $C = \{(x, y) \in \mathbb{C}^2 : y^2 = x^3\}$ is singular at $(0, 0)$. In particular, maximal ideals are in 1-to-1 correspondence with points on C: if, say, $(\alpha, \beta) \in C$, then $\langle x - \alpha, y - \beta \rangle$ is a maximal ideal. Thus we observe that every maximal ideal of A_3 is generated by two elements at least, compared to other rings A_i.

A_5 and A_6. The ring A_5 is isomorphic to the ring of truncated polynomials $\mathbb{C}[x]_{<6}$. The dimension of A_5 (as $\mathbb{C}$-vector space with basis $\{1, x, \dots, x^5\}$) is 6, so it is clearly not isomorphic to anything else besides probably A_6 (which also has a finite dimension of 6 with the basis given by $\{1, x, y, y^2, xy, xy^2\}$). However, it is not isomorphic to A_5 since maximal ideal $\mathfrak{m}$ of A_5 is generated by a single element x, while in A_6 the ideal $\mathfrak{m}/\mathfrak{m}^2$ (if we vanish all quadratic terms x^2, xy, y^2) is a vector space over two elements x and y.

A_7, A_8, and A_9. We consider nilradicals of these rings. Note that $\mathrm{nil}(A_7) = 0$,

[*]Indeed, denote by $S = \{2i + 3j : i, j \in \mathbb{Z}_{\geq 0}\}$. Clearly, $2 \in S$. Also note that if $y \in S$, then $y + 1 \in S$: if $y = 2i_0 + 3j_0$, then $y + 1 = 2(i_0 - 1) + 3(j_0 + 1)$. Thus, $S = \mathbb{Z}_{\geq 0} \setminus \{1\}$.

[†]Credits to this mathstackexchange thread.

$\mathrm{nil}(A_8) = \langle xy \rangle$ and $\mathrm{nil}(A_9) = \langle y \rangle$. We see immediately that A_7 and A_8 are not isomorphic because A_7 is reduced while A_8 is not. Now compare quotients $A_8/\mathrm{nil}(A_8)$, and $A_9/\mathrm{nil}(A_9)$. Notice that $A_8/\mathrm{nil}(A_8) = \mathbb{C}[x,y]/\langle xy^2 \rangle/\langle xy \rangle \cong \mathbb{C}[x,y]/\langle xy \rangle$. In turn, $A_9/\mathrm{nil}(A_9) = \mathbb{C}[x,y]/\langle xy, y^2 \rangle/\langle y \rangle \cong \mathbb{C}[x,y]/\langle xy, y^2, y \rangle \cong \mathbb{C}[x,y]/\langle xy, y \rangle \cong \mathbb{C}[x]$. Since $\mathbb{C}[x,y]/\langle xy \rangle$ is not an integral domain (take $x, y \in \mathbb{C}[x,y]/\langle xy \rangle$ and note that their product is 0), $\mathbb{C}[x] \ncong \mathbb{C}[x,y]/\langle xy \rangle$ and thus A_8 is not isomorphic to A_9.

Problem (Exercise 2)**.** *Let $X = Spec(A)$ and $x, y \in X$. Show that:*

(a) x is a closed point (i.e. $\{x\} \subseteq X$ is a closed subset) iff the corresponding prime ideal $\mathfrak{p}_x \subseteq A$ is maximal;

(b) the closure $\overline{\{x\}} = V(\mathfrak{p}_x)$;

(c) $y \in \overline{\{x\}}$ iff $\mathfrak{p}_x \subseteq \mathfrak{p}_y$.

Solution.

Part (a). Suppose $\mathfrak{p}_x$ is maximal. Then by definition there is no proper prime ideal containing $\mathfrak{p}_x$. Therefore, $V(\mathfrak{p}_x) = \{\mathfrak{p}_x\} = \{x\}$, and so $\{x\}$ is in fact closed. Now suppose $\{x\}$ is closed, then $\{x\} = V(\mathfrak{p}_x)$. Suppose $\mathfrak{p}_x$ is not maximal, meaning there exists some $\mathfrak{p}_y$ between $\mathfrak{p}_x$ and A. But then $\mathfrak{p}_y \in V(\mathfrak{p}_x)$ and so $V(\mathfrak{p}_x)$ contains at least two elements. This is a contradiction. □

Part (b). We prove the auxilary proposition.

Proposition (Atiyah-Macdonald 1.11, (ii))**.** *Let $\mathfrak{a}_1, \ldots, \mathfrak{a}_n$ be ideals and $\mathfrak{p} \supseteq \bigcap_{i=1}^n \mathfrak{a}_i$. Then, $\mathfrak{p} \supseteq \mathfrak{a}_i$ for some i. In particular, if $\mathfrak{p} = \bigcap_{i=1}^n \mathfrak{a}_i$, then $\mathfrak{p} = \mathfrak{a}_i$.*

Proof. We prove the contrapositive statement. Suppose $\mathfrak{p} \nsubseteq \mathfrak{a}_i$ for all i. Then, there exist $x_i \in \mathfrak{a}_i \setminus \mathfrak{p}$. Now note $\prod_{i=1}^n x_i \in \prod_{i=1}^n \mathfrak{a}_i \subseteq \bigcap_{i=1}^n \mathfrak{a}_i$, but $\prod_{i=1}^n x_i \notin \mathfrak{p}$ since $\mathfrak{p}$ is prime. Hence $\mathfrak{p} \ngeq \bigcap_{i=1}^n \mathfrak{a}_i$. □

Now, *part (b)* is an immediate consequence. By definition, $\overline{\{x\}} = \bigcap_{x \in V(I)} V(I)$. Since closure is a closed set, $\overline{\{x\}}$ is a prime ideal, so $\{x\} = V(I)$ for some ideal I. This is exactly what we denote by $\mathfrak{p}_x$. □

Part (c). If $y \in \overline{\{x\}}$, then $\{y\} \subseteq V(\mathfrak{p}_x)$. Take closure of both sides. One then has $\overline{V(\mathfrak{p}_y)} \subseteq V(\mathfrak{p}_x)$, implying $\mathfrak{p}_x \subseteq \mathfrak{p}_y$. We can do the same in the other direction. If $\mathfrak{p}_x \subseteq \mathfrak{p}_y$, then $\overline{\{y\}} = V(\mathfrak{p}_y) \subseteq V(\mathfrak{p}_x) = \overline{\{x\}}$, which implies $y \in \overline{\{x\}}$. □

Problem (Exercise 3)**.** *We say that topogical space X is **irreducible** if for every pair of closed subsets $Y_0, Y_1 \subseteq X$ such that $X = Y_0 \cup Y_1$, we have $X = Y_0$ or $X = Y_1$. Show that Spec(A) is irreducible if and only if the nilradical $\sqrt{0} \subseteq A$ is a prime ideal.*

Proof. Recall that set Y is closed if $Y = V(I)$. Now, if $\text{Spec}(A) = Y$, then $V(I) = \text{Spec}(A)$ or equivalently $I \subseteq \bigcap_{\mathfrak{p} \in \text{Spec}(A)} \mathfrak{p}$. According to the previous lectures, $\bigcap_{\mathfrak{p} \in \text{Spec}(A)} \mathfrak{p} = \sqrt{0}$. Therefore, $V(I) = \text{Spec}(A)$ iff $I \subseteq \sqrt{0}$.

Now suppose Spec(A) is irreducible, meaning that if $\text{Spec}(A) = V(I) \cup V(J) = V(IJ)$, then either $\text{Spec}(A) = V(I)$ or $\text{Spec}(A) = V(J)$. This is equivalent to saying that whenever $IJ \subseteq \sqrt{0}$ then $I \subseteq \sqrt{0}$ or $J \subseteq \sqrt{0}$, but this exactly means $\sqrt{0}$ is a prime ideal.

Now assume $\sqrt{0} \subseteq A$ is a prime ideal and assume $\text{Spec}(A) = V(I) \cup V(J) = V(IJ)$. Thus, $IJ \subseteq \sqrt{0}$, but since $\sqrt{0}$ is prime, this implies either $I \subseteq \sqrt{0}$ or $J \subseteq \sqrt{0}$, which is equivalent to saying $V(I) = \text{Spec}(A)$ or $V(J) = \text{Spec}(A)$. Thus, A is irreducible. □

Problem (Exercise 4)**.** *Let X be a topological space. Show the following:*

(a) If $Y \subseteq X$ is an irreducible subspace, then its closure $\overline{Y}$ is irreducible.

(b) Every irreducible subspace of X is contained in a maximal irreducible subspace.

*(c) The maximal irreducible subspaces of X are closed and cover X. They are called the **irreducible components** of X. What are the irreducible components of a Hausdorff space?*

(d) If A is a ring and $X = \text{Spec}(A)$, then the irreducible components of X are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal of A.

Solution.

Part (a). Suppose that $\overline{Y}$ is reducible. Thus, there exist closed subsets $G_0, G_1 \subsetneq \overline{Y}$ with $\overline{Y} = G_0 \cup G_1$. Since G_0 and G_1 are closed in $\overline{Y}$, there exist closed sets $X_0, X_1 \subseteq X$ such that $G_0 = X_0 \cap \overline{Y}$ and $G_1 = X_1 \cap \overline{Y}$. Then,

$$Y \subseteq \overline{Y} = G_0 \cup G_1 = (X_0 \cap \overline{Y}) \cup (X_1 \cap \overline{Y})$$

Therefore, $Y = (Y \cap X_0) \cup (Y \cap X_1)$, where $Y \cap X_i$ are closed in Y. Since Y is irreducible, one of these equals Y. Suppose $Y \cap X_0 = Y$ or $Y \subseteq X_0$. But then $\overline{Y} \subseteq \overline{X}_0 = X_0$ so $\overline{Y} = \overline{Y} \cap A = G_0$. This is a contradiction.

Part (b). Suppose $\{A_\alpha\}_{\alpha \in I}$ be a chain of irreducible subsets of X. We shall show that their union $\mathsf{A} := \bigcup_{\alpha \in I} A_\alpha$ is irreducible. Suppose it is not. Then, there

exist two closed sets $U, V \subsetneq \mathsf{A}$ such that $\mathsf{A} = U \cup V$. Clearly, there exist two indices, say β, γ, such that $U \cap A_\beta$ and $V \cap A_\gamma$ are non-empty. Since $\{A_\alpha\}_{\alpha \in I}$ is a chain, either $A_\beta \subseteq A_\gamma$ or $A_\gamma \subseteq A_\beta$. Suppose the former. Then, $U \cap A_\gamma$ and $V \cap A_\gamma$ are both closed in A_γ, so either $U \cap A_\gamma = A_\gamma$ or $V \cap A_\gamma = A_\gamma$ since A_γ is irreducible. This implies that either $A_\gamma \subseteq U$ or $A_\gamma \subseteq V$. However, this is a contradiction: if $A_\gamma \subseteq U$, then $V \cap A_\gamma = \emptyset$, hence the assumption of A being irreducible is impossible.

Now we simply apply the *Zorn's lemma*: for each $Y \subseteq X$, consider the family of irreducible sets that contain Y. The union of such sets, as shown above, is clearly its maximal element.

Part (c). From *part (b)*, every irreducible Y is contained in the corresponding maximal irreducible subspace Y. By *part (a)*, we know that $\overline{\mathsf{Y}}$ is also irreducible. But since $\mathsf{Y} \subseteq \overline{\mathsf{Y}}$ and Y is maximal, this forces $\mathsf{Y} = \overline{\mathsf{Y}}$, which means Y is closed.

Now we show that these maximal subspaces cover X. Take some $\{x\} \subseteq X$. Clearly, $\{x\}$ is irreducible. From *part (b)*, there exists some maximal irreducible space V_x such that $\{x\} \subseteq V_x$. Therefore, $\bigcup_{x \in X} V_x$ covers X.

For Hausdorff space, irreducible components are simply singleton sets $\{x\}$.

Part (d). Following Exercise 3, we know (with minor adjustments) that $V(I)$ is irreducible iff $\sqrt{I}$ is a prime ideal. Since $V(I) = V(\sqrt{I})$, this implies that a closed subset of $\mathrm{Spec}(A)$ is closed iff it is of form $V(\mathfrak{p})$ for some prime $\mathfrak{p} \subseteq A$.

At this point, we are almost done. First note that $V(\mathfrak{p})$ is an irreducible component if $\mathfrak{p}$ is minimal. Firstly, it is clearly irreducible. Suppose U is an irreducible closed set such that $V(\mathfrak{p}) \subseteq U = V(\mathfrak{q})$. Then, $\mathfrak{q} \subseteq \mathfrak{p}$, but since $\mathfrak{p}$ is minimal, $\mathfrak{q} = \mathfrak{p}$. Thus $V(\mathfrak{p})$ is maximal.

Finally, we have to show that every irreducible component is of form $V(\mathfrak{p})$ for minimal $\mathfrak{p}$. But since every irreducible component is closed, it is of form $V(\mathfrak{p})$ for some prime $\mathfrak{p}$. For it to be maximal, as shown above, $\mathfrak{p}$ must be minimal. □

Problem (Exercise 5)**.** *Let X be a topological space. We say that X is*

1. ***sober*** *if every irreducible closed subset $Y \subseteq X$ has a unique generic point, i.e. there exists a unique $\eta_Y \in Y$ such that $Y = \overline{\{\eta_Y\}}$;*
2. ***quasi-compact*** *(qc) if every open cover has a finite subcover;*
3. ***spectral*** *if it is sober, qc, if the intersection of every two qc open subsets of X is qc, and if its qc open subsets form a base for the topology.*

Let A be a commutative ring. Prove that $\mathrm{Spec}(A)$ is a spectral space.

Proof. $\mathrm{Spec}(A)$ is clearly sober: every irreducible closed subset Y is of form $Y = V(\mathfrak{p})$, as we have shown in Exercise 3 (and used in Exercise 4). But in Exercise 2 we have also shown that $V(\mathfrak{p}) = \overline{\{\eta_Y\}}$ for some $\eta_Y \in Y$, so $Y = \overline{\{\eta_Y\}}$.

Now recall that open sets in $\mathrm{Spec}(A)$ are $D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) : f \notin \mathfrak{p}\}$. Suppose $\mathrm{Spec}(A) = \bigcup_{\alpha\in I} D(f_\alpha)$ is an open cover.

Now, consider the ideal $\mathfrak{a} := \langle f_\alpha\rangle_{\alpha\in I} \subseteq A$. Assume $\mathfrak{a} \subsetneq A$, then $\mathfrak{a}$ is contained in some maximal ideal, and thus in some prime ideal $\mathfrak{p}$. Note that for each f_α, $f_\alpha \in \mathfrak{a} \subseteq \mathfrak{p}$, so $\mathfrak{p} \notin D(f_\alpha)$, and thus $\mathfrak{p} \notin \bigcup_{\alpha\in I} D(f_\alpha)$. But $\bigcup_{\alpha\in I} D(f_\alpha)$ covers the whole $\mathrm{Spec}(A)$, so we reached the contradiction. Thus, $\langle f_\alpha\rangle_{\alpha\in I} = A$, so in particular there is a finite subset $J \subseteq I$ such that $\sum_{\alpha\in J} c_\alpha f_\alpha = 1$. It is thus easy to see that $\bigcup_{\alpha\in J} D(f_\alpha) = \mathrm{Spec}(A)$ is a finite open subcover.

Two left properties are then trivial: note that $D(f)$ is open, qc and forms a basis for topology. And since $D(f) \cap D(g) = D(fg)$, it follows that intersection of two qc open sets is again a qc open set. □